Toolkit: Chemistry of the Earth

Practical Seven: Melting and the Rare Earth Elements (REE)

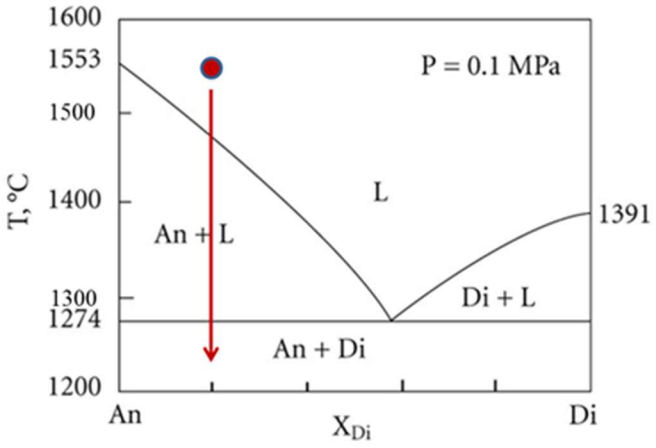
Objectives:

- Learn how to interpret partial melting of a rock as shown in a phase diagram
- Understand why the Rare Earth Elements (REE) are powerful trace elements that can be used to interpret geochemical processes:
    - Investigate the behaviour of REE in different minerals
    - Use the Rare Earth Elements (REE) to interpret melting processes in the mantle

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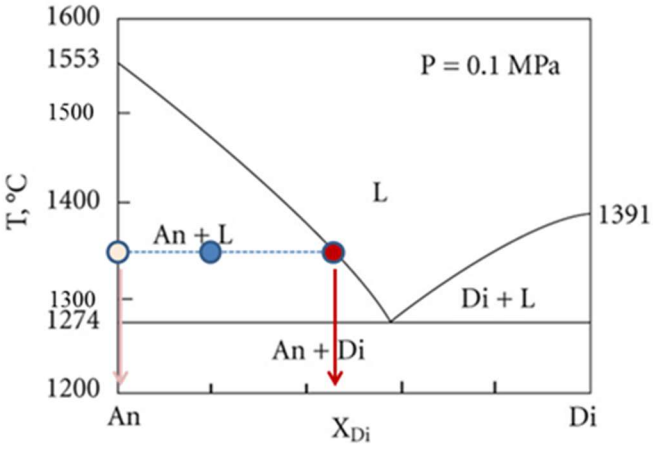
FEEDBACK

How to read a binary phase diagram: Example, Anorthite – Diopside



Liquid only

Composition of liquid: $An_{80}Di_{20}$



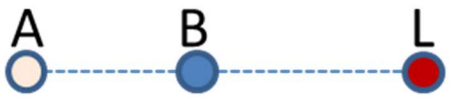
Liquid + Anorthite

Composition of liquid: $An_{55}Di_{45}$

Proportions:

% Liquid = AB / AL

% Anorthite = BL / AL



A. Understand how partial melting of a rock is shown in a phase diagram

The phase diagram below shows the melting relationships for diopside and anorthite. (Note that the x axis is reversed compared with the version on the previous page.) There is one region where only one phase is stable: Liquid. There are three regions where two phases are stable: these are Diopside + Anorthite; Diopside + Liquid; Anorthite + Liquid. *Identify these different regions* on the phase diagram.

On the phase diagram, track the melting behaviour of a mixture of 70% anorthite, 30% diopside, by showing the stable phases at 1250, 1300, 1400 and 1500 °C. Complete the table below to show the *proportions* of phases in two-phase regions, and the composition of the liquid at each temperature. In the boxes, draw a sketch of what the system looks like at each temperature.

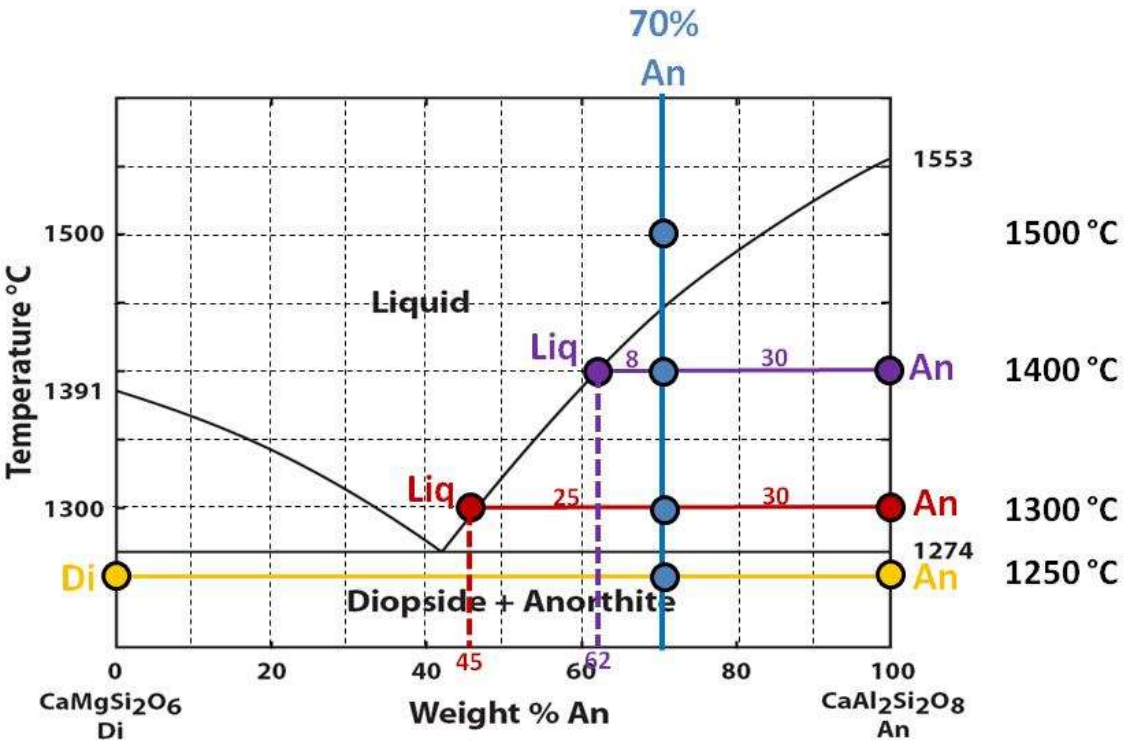
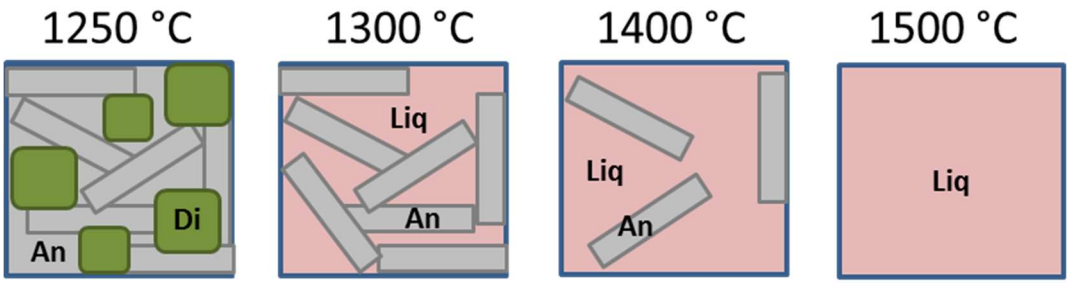


Table 1: Melting a mixture of 70% anorthite, 30% diopside			
Temperature	Phases present	Proportions of phases	Composition of liquid
1250 °C	Anorthite & Diopside	70% An, 30% Di	n/a
1300 °C	Anorthite & Liquid	55% Liq, 45% An	$An_{45}Di_{55}$
1400 °C	Anorthite & Liquid	79% Liq, 21% An	$An_{62}Di_{38}$
1500 °C	Liquid	100% Liq	$An_{70}Di_{30}$



a) For the composition considered above (70% anorthite), what is the temperature when the first melt appears (the eutectic temperature)?

The eutectic temperature is 1274 °C. For any mixture of anorthite + diopside, this is the temperature when the first melt appears.

b) For the composition considered above (70% anorthite), what is the liquid composition of the first melt?

The composition of the eutectic is $An_{42}Di_{58}$. For any mixture of anorthite + diopside, this is the composition of the first melt.

c) For the composition considered above (70% anorthite), what is the temperature of the point of complete melting?

Complete melting for this composition occurs at 1450 °C. This is the point where the bulk composition crosses the phase boundary between the field of Liquid and the field of Liquid+Anorthite. The point of complete melting depends on the bulk composition we are considering.

d) For the composition considered above (70% anorthite), how does the composition of the liquid change as temperature increases, and why?

The first liquid composition (eutectic) is An_{42}. At this point all of the diopside and some of the anorthite melts. As we increase the temperature, more of the anorthite melts and the liquid composition gradually approaches the bulk composition (An_{45} at 1300 °C, An_{62} at 1400 °C, An_{70} at 1450 °C).

e) For the composition considered above (70% anorthite), what is the total temperature range where the two-phase assemblage anorthite + melt is present?

The system consists of Anorthite + Liquid starting just above the eutectic (1275 °C) up to just below the temperature of total melting (1450 °C). In other words, the system consists of Anorthite + Liquid over a temperature range of 175 °C.

f) How does the temperature range of the two-phase assemblage vary if we change the proportions of anorthite and diopside in the rock we are considering (e.g. 90% anorthite, 50% anorthite, 10% anorthite)?

For 90% Anorthite, the temperature range is about 250 °C.

For 50% Anorthite, the temperature range is about 50 °C.

For 10% Anorthite, the temperature range is about 100 °C – for this composition, the two-phase assemblage is Diopside + Liquid.

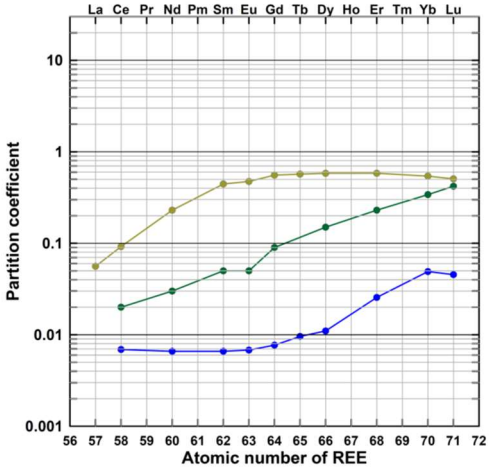
Overall, there is a big difference in the range of temperatures where partial melting occurs, depending on the composition being considered.

B. Investigate the behaviour of REE in different minerals.

1. The Table below gives mineral / melt REE partition coefficients for plagioclase, pyroxene, olivine and garnet, for equilibrium with basaltic liquids. (Values are from Rollinson: Using Geochemical Data.)

Table 2: REE partition coefficients, basaltic liquids						
	Z	Olivine	Orthopyroxene	Clinopyroxene	Plagioclase	Garnet
La	57			0.056	0.148	0.026
Ce	58	0.007	0.020	0.092	0.082	0.051
Nd	60	0.007	0.030	0.230	0.055	
Sm	62	0.007	0.050	0.445	0.039	0.600
Eu	63	0.007	0.050	0.474	1.126	1.000
Gd	64	0.008	0.090	0.556	0.031	2.100
Tb	65	0.010		0.570		
Dy	66	0.011	0.150	0.582	0.023	4.100
Ho	67					
Er	68	0.026	0.230	0.583	0.020	6.560
Yb	70	0.049	0.340	0.542	0.023	11.500
Lu	71	0.045	0.420	0.506	0.019	11.900

The graph below shows REE partition coefficients for olivine, orthopyroxene and clinopyroxene, plotted as a function of atomic number.



The formulae of these three minerals are:

Olivine $(Mg,Fe)_2SiO_4$

Orthopyroxene $(Mg,Fe)_2Si_2O_6$

Clinopyroxene $(Mg,Fe)CaSi_2O_6$

a) Are the REE compatible or incompatible elements in olivine and pyroxene? – explain.

The REE are incompatible in olivine and pyroxene – all the partition coefficients are less than 1.

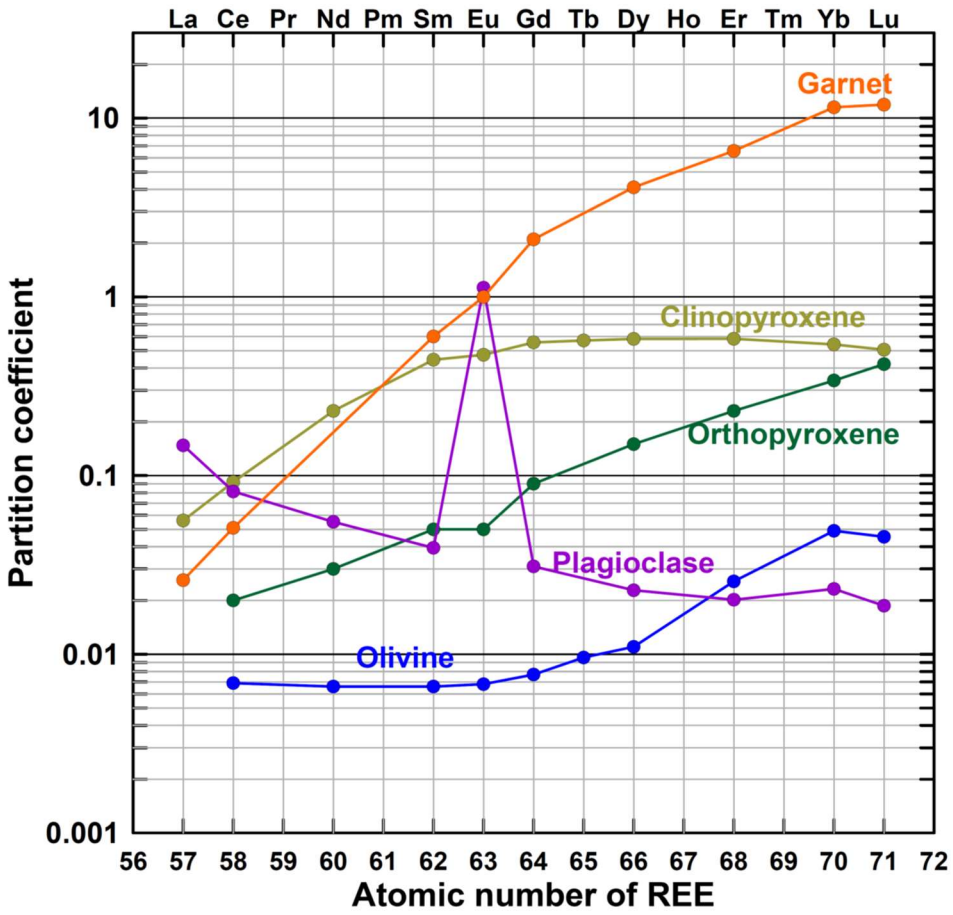
b) Compare olivine and orthopyroxene. Their mineral formulae are very similar. What could be the reason for their different partition coefficients?

Olivine and orthopyroxene have different arrangements of atoms. In olivine, silica tetrahedra are isolated. In pyroxene, silica tetrahedra are arranged in chains. The different silicate structures mean that there are different sized sites for cations, and the pyroxene structure is able to incorporate REE a little more than olivine.

c) Compare orthopyroxene and clinopyroxene. What is the most significant chemical difference between the two types of pyroxene, and why might this have an effect on the partition coefficients?

The difference is that clinopyroxene contains significant amounts of Ca. Ca^{2+} is a large ion. The presence of Ca opens up the structure and allows for more incorporation of large REE ions.

d) This graph shows the same data as above. Now add the partition coefficients for plagioclase and garnet to the graph.



e) Are the REE compatible or incompatible elements in garnet?

A mixture. The light REE (LREE) are incompatible (D < 1) and the heavy REE (HREE) are compatible (D > 1).

f) Are the REE compatible or incompatible elements in plagioclase?

Essentially all the REE are incompatible (D < 1), although Eu is slightly compatible (D just a little >1).

g) Why doesn't Eu fit the main trend for plagioclase?

Eu prefers to occur as Eu^{2+} whereas all the other REE occur as 3+ cations. This means it behaves differently. Eu^{2+} can substitute easily into the Ca^{2+} site in plagioclase because it has a similar size and charge, so it has a much higher partition coefficient than the other REEs.

h) The difference in behaviour of Eu is called a Eu anomaly. For plagioclase, this is described as a positive Eu anomaly. Which minerals show a negative Eu anomaly?

Orthopyroxene and clinopyroxene show negative Eu anomalies, although these are much less pronounced than the positive Eu anomaly in plagioclase.

i) In some minerals, La is more compatible than Lu. In others, the reverse is true. Which minerals show which patterns?

La is more compatible than Lu in plagioclase; Lu is more compatible than La in olivine, orthopyroxene, clinopyroxene, garnet.

C. Use the Rare Earth Elements (REE) to understand melting processes in the Earth's mantle

We are going to consider the composition of a liquid produced by melting of the mantle. As an approximation, we can calculate a "Bulk partition coefficient" that describes the composition of a liquid in equilibrium with a mixture of minerals.

Bulk partition coefficient = Bulk D = $x_1D_1 + x_2D_2 + x_3D_3 + \ldots\ldots$

where x_1 = fraction of mineral 1 in the rock.

For example, if we consider Ti in a rock that contains 50% olivine, 30% orthopyroxene and 20% clinopyroxene,

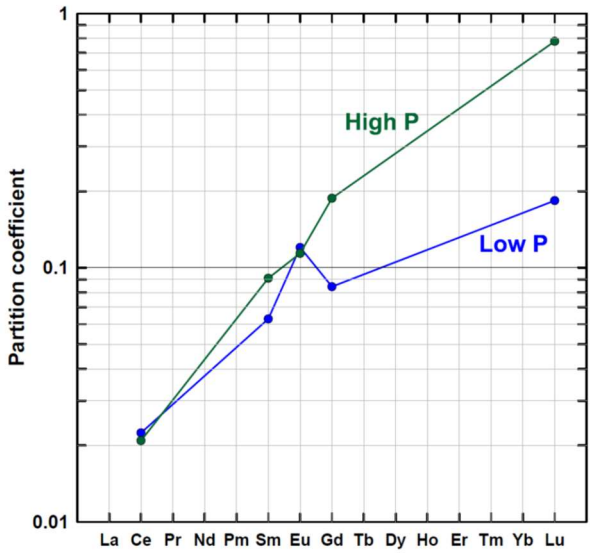
Bulk D_{Ti} = $(0.5 * {}^{olv}D_{Ti}) + (0.3 * {}^{opx}D_{Ti}) + (0.2 * {}^{cpx}D_{Ti})$

If we produce a melt in the mantle, it will carry a signature of the minerals that were present in the source region. This can be an important indicator of the depth at which melting occurred. For example, mantle peridotite contains plagioclase at shallow depths (low pressures) but at greater depths (high pressures), plagioclase is not stable and garnet is preferred. If we measure the REE abundances in an erupted melt, we can learn about the depth of melt formation.

a) The table below shows the bulk partition coefficients for selected REE, for a rock that contains olivine, orthopyroxene, clinopyroxene and either plagioclase or garnet, in the proportions shown. The values for the low pressure case are given. Calculate the values for high pressure melting.

Table 3: Bulk partition coefficients for mantle peridotite 60% olivine, 25% opx, 10% cpx, 5% plagioclase or garnet		
	Low pressure: Plagioclase	High pressure: Garnet
Ce	0.022	0.021
Sm	0.063	0.091
Eu	0.120	0.114
Gd	0.084	0.188
Lu	0.184	0.778

Bulk D_{Ce} = $(0.6 \times {}^{olv}D_{Ce}) + (0.25 \times {}^{opx}D_{Ce}) + (0.1 \times {}^{cpx}D_{Ce}) + (0.05 \times {}^{gnt}D_{Ce})$



Plot the partition coefficients for the two cases of melting on this graph.

In general, are REE compatible or incompatible elements in the mantle?

They are incompatible elements, because the partition coefficients are less than one. This means they prefer to be in the melt. A small amount of melt produced in the mantle has a high concentration of REE.

b) Use the partition coefficients you calculated above to determine the REE abundance pattern for a melt derived from the mantle at low pressure and at high pressure.

We will consider that the mantle has the same REE concentrations as chondrites (CI chondrites). Hence, the normalized REE abundances in the mantle (Mantle/CI chondrite) all have a value of 1. To calculate the composition of the liquid, simply use the partition coefficient formula,

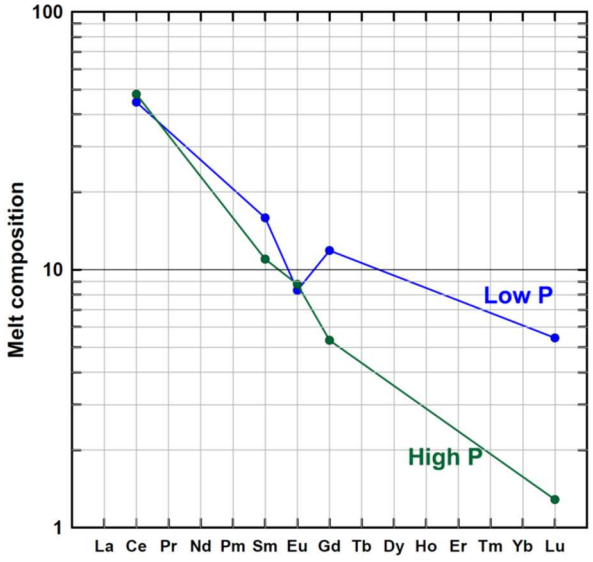
$$D_{bulk} = \frac{\text{(Abundance of element in Mantle)}}{\text{(Abundance of element in Liquid)}}$$

- which in this case becomes: (Abundance of element in Liquid) = $1/D_{bulk}$

Table 4: Composition of a liquid derived from the mantle (relative to CI chondrite)

	Mantle abundance / CI	Low pressure / CI	High pressure /CI
Ce	1	44.61	47.87
Sm	1	15.89	10.99
Eu	1	8.32	8.77
Gd	1	11.87	5.33
Lu	1	5.44	1.29

On the graph below, compare the REE abundance patterns you would expect for melts produced at low pressure and high pressure.



What characteristics of the REE pattern would you look for in a basalt, in order to argue that a melt was produced at high vs low pressure in the mantle?

A melt produced at high pressure would have garnet in its source. This means it would have a REE pattern with a steep slope and a small positive Eu anomaly. In contrast, a melt produced by melting at low pressure would have plagioclase in its source. This melt would have a REE pattern with a shallower slope and a significant negative Eu anomaly.

Have you noticed that the melt composition is the inverse (mirror image) of the partition coefficient? Also, notice that REE contents of the melts are high compared with the (unmelted) mantle.